Order of operations, and simple equations

On the cheat sheet: **side 1 → ORDER OF OPERATIONS, AND EQUATIONS**

Work it out (2 marks)

`5 + 2³ ÷ 4 − 1`

1 Do NOT go left to right. There is an order, and powers go first. The little 3 means 2 multiplied by itself three times.

`2³ = 2×2×2 = 8`

Cheat sheet: **ORDER OF OPERATIONS** box, the BODMAS line at the top.

2 Now the sum reads like this. Still do not go left to right.

`5 + 8 ÷ 4 − 1`

3 Divide and multiply come before add and subtract, so the 8 ÷ 4 happens next, on its own.

`8 ÷ 4 = 2`

4 Only NOW do the adding and subtracting — and for those you DO go left to right.

`5 + 2 − 1 = 6`

Doing 5 + 8 first gives 13 ÷ 4. That is the whole trap.

ANSWER 6

Solve for t (2 marks)

`7t − (3 − t) = 2`

1 There is a minus sitting in front of a bracket. That minus belongs to EVERYTHING inside, not just the first thing.

`both signs inside flip`

Cheat sheet: **ORDER OF OPERATIONS** table, row 7t − (3 − t).

2 So the 3 becomes minus 3, and the minus t becomes PLUS t.

`7t − 3 + t = 2`

3 Collect the t. Seven of them plus one more is eight.

`8t − 3 = 2`

4 The minus 3 went on last, so it comes off first. Add 3 to both sides, then divide both sides by 8.

`8t = 5 → t = 5/8 = 0.63`

Undo in reverse order — last thing on, first thing off.

ANSWER `t = 5/8 = 0.63`

B	brackets	5 + 2³ ÷ 4 − 1	none here, so move on
O	orders (powers)	5 + 2³ ÷ 4 − 1	$2^3 = 8$
DM	divide, multiply	5 + 8 ÷ 4 − 1	8 ÷ 4 = 2
AS	add, subtract	5 + 2 − 1	left to right → 6

The same sum, peeled one layer at a time. The underlined bit is the only part you touch on that pass.

Spotting it in three seconds

If the question looks like	Your first move
`a power or a root anywhere`	do that bit first, on its own
`− in front of a bracket`	flip every sign inside
`fraction with x on the bottom`	multiply both sides by the bottom
`x on both sides`	get all the x on one side first

Watch out. Adding and subtracting are equal in rank — do them left to right. Same for multiply and divide. Only the four LEVELS are ordered.

Two equations at once, and inequalities

On the cheat sheet: **side 1 → ORDER OF OPERATIONS, AND EQUATIONS, lower rows**

Solve both together (3 marks)

`5x + 2y = 13    and    x + 2y = 9`

1. Look at the two lines before doing anything. Is any part IDENTICAL in both? Yes — both contain exactly 2y.

   `both contain 2y`

   Cheat sheet: row **5x+2y=13, x+2y=9** — a term matches, so subtract.

2. When a term is identical in both, taking one whole line away from the other makes it vanish. Nothing needs preparing.

   `(5x + 2y) − (x + 2y) = 13 − 9`

3. The 2y cancels itself out and only x is left.

   `4x = 4  →  x = 1`

4. Put that x back into whichever line looks easier, and get y.

   `1 + 2y = 9  →  y = 4`

5. Check in the OTHER line — the one you did not use.

   `5(1) + 2(4) = 13 ✓`

ANSWER `x = 1, y = 4`

Solve the inequality (2 marks)

`−5c − 7 > 8`

1. Treat it exactly like a normal equation, right up until the last step.

   `add 7 to both sides`

2. That gives this. Nothing strange has happened yet.

   `−5c > 15`

3. **Here is the whole topic.** To get c alone you divide by MINUS 5, and dividing by a negative turns the arrow round.

   `greater-than becomes less-than`

   Cheat sheet: row **−5c − 7 > 8**, in red.

4. So the answer flips direction, and 15 divided by minus 5 is minus 3.

   `c < −3`

5. Test a number that ought to work, say c = −4.

   `−5(−4) − 7 = 13, and 13 > 8 ✓`

ANSWER `c < −3`

Two equations: kill the part that already matches

5x + $\boxed{2y}$ = 13

x + $\boxed{2y}$ = 9

− ⟶ **4x = 4**

the 2y is gone

Inequality: the arrow turns round when you divide by a minus

c < −3

The circle is hollow because −3 itself is not allowed — only everything to the left of it.

Left: the matching part is what gets killed. Right: the hollow circle means −3 itself is not included.

Spotting it in three seconds

If the question looks like	Your first move
`a term matches exactly`	subtract one line from the other
`signs opposite, e.g. +2y and −2y`	ADD the two lines instead
`nothing matches`	multiply one line up until something does
`one line already says x = ...`	substitute it into the other

Watch out. The flip only happens for a NEGATIVE multiply or divide. Adding or taking away a negative number changes nothing about the arrow.

Quadratics

On the cheat sheet: **side 1 → QUADRATICS**

Find the roots — 3 marks

`y = 6x² + 14x − 12`

1. "Roots" means where the curve crosses the ground, and on the ground y is nothing. So set it to zero.

   `6x² + 14x − 12 = 0`

2. Write the three numbers down BEFORE touching the formula, signs included.

   `a = 6    b = 14    c = −12`

   Cheat sheet: row **roots, does not factorise** — name a, b, c first.

3. Do the bit under the square root on its own. Careful: $4\times6\times(-12)$ is minus 288, and taking away a minus ADDS it.

   `b² − 4ac = 196 + 288 = 484`

4. Square root it. It comes out exact, which is a sign you are on track.

   `√484 = 22`

5. The rest of the formula is on your OFFICIAL sheet. The ± means do the sum twice, once each way — that is why there are two answers.

   `x = (−14 ± 22) ÷ 12`

ANSWER `x = 2/3   or   x = −3`

Completed square form — 4 marks

`y = 2x² + 8x + 15    as    p(x + q)² + r`

1. There is a number in front of the x^2. Take it out of the FIRST TWO terms only, and leave the 15 alone outside.

   `y = 2(x² + 4x) + 15`

   Cheat sheet: row **$p(x+q)^2+r$, $a \neq 1$**, in red.

2. Inside the bracket, halve the middle number. Half of 4 is 2, so the bracket is (x + 2) squared — but that is too big by 2 squared, so take 4 back off.

   `y = 2((x + 2)² − 4) + 15`

3. **This is where the mark goes.** The 2 outside multiplies the −4 as well as the bracket, so it becomes −8.

   `y = 2(x + 2)² − 8 + 15`

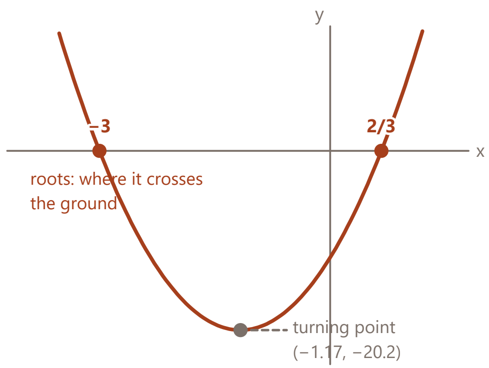
4. Tidy the two loose numbers on the end.

   `y = 2(x + 2)² + 7`

5. The turning point falls straight out: flip the sign of the number inside the bracket, keep the sign of the one outside.

   `turning point (−2, 7)`

ANSWER `q = 2, r = 7, turning point (−2, 7)`



The curve for the left-hand question, drawn from the actual numbers. Roots on the ground, turning point at the bottom.

What $b^2 - 4ac$ tells you before you solve

`> 0`	two roots	it cuts the axis twice
`= 0`	one root	it just touches
`< 0`	no roots	it never reaches the axis

a positive → smile → turning point is the LOWEST
a negative → frown → turning point is the HIGHEST

`p(x + q)² + r → turning point (−q, r)`
flip the sign of q. Keep the sign of r.

Spotting it in three seconds

If the question looks like	Your first move
`roots / solve / where it cuts`	set y = 0, then the formula
`p(x+q)² + r`	complete the square
`maximum or minimum value`	complete the square, read off r
`how many roots?`	work out $b^2 - 4ac$ and look at its sign

Watch out. The quadratic formula IS on your official sheet. What is not on it: what the discriminant sign means, and that a positive a makes a smile.

Straight lines

On the cheat sheet: **side 1 → STRAIGHT LINES AND COORDINATE GEOMETRY**

Find the equation (3 marks)

`through (2, 4) and (5, -2)`

1. You have two points and no gradient, so the gradient is the first job. It is how far it climbs divided by how far it goes across.

   $m = (-2 - 4) \div (5 - 2)$

   Cheat sheet: row **two points → gradient**.

2. Careful with the top: minus two take away four is minus SIX.

   $m = -6 \div 3 = -2$

3. Now pick either point — genuinely either — and use this shape.

   $y - y_1 = m(x - x_1)$

   Cheat sheet: row **gradient + a point**. NOT on your official sheet.

4. Drop the numbers into their slots and open the bracket. Minus two times minus two is plus four.

   $y - 4 = -2(x - 2) \rightarrow y - 4 = -2x + 4$

5. Tidy up, then check with the point you did NOT use: put $x = 5$ in and see whether you get -2.

   $y = -2x + 8 \quad -10 + 8 = -2$ ✓

ANSWER $y = -2x + 8$

Which of these are parallel? (2 marks)

`to y = 3x + 4 · A 3y=12x+6 B 4y=12x+5 C y=4x+4 D y=6x+8 E 6x−2y=10`

1. Parallel means the same steepness and nothing else. Steepness is the number in front of x — but only once y is on its own.

   `the line to match has m = 3`

   Cheat sheet: row **parallel?** — rearrange every option first.

2. A: divide every term by 3. That gives $m = 4$. Not it.

   $y = 4x + 2$

3. B: divide every term by 4. That gives $m = 3$. **Yes.**

   $y = 3x + 1.25$

4. C is already tidy with $m = 4$, D with $m = 6$. Neither.

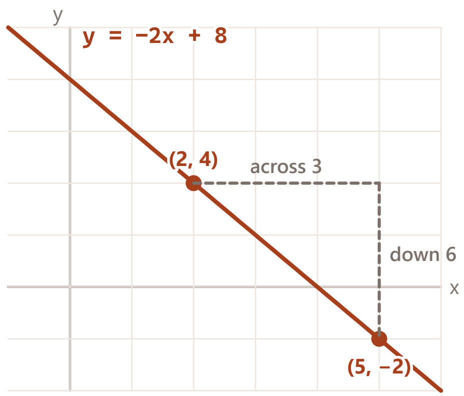
   `no, and no`

5. E is the one worth the time, because it does not even look like a line. Move the 6x across, then divide BOTH sides by minus 2.

   $6x - 2y = 10 \rightarrow y = 3x - 5$ **yes**

   It says select ALL. Stopping at B loses the mark even though B is right.

ANSWER B and E



The left-hand line, with the drop of 6 over a run of 3 drawn in. Going down as you go right is what a negative gradient looks like.

gradient m = rise ÷ run = $-6 \div 3 = -2$

Going DOWN as you go right means m is negative.

parallel same m

never meet, so only the c changes

perpendicular $m_1 \times m_2 = -1$

flip it over, change the sign:
$m = -2 \rightarrow$ perpendicular $1/2$

midpoint = average the x, average the y
distance = $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ — both on your official sheet

Spotting it in three seconds

If the question looks like	Your first move
two points given	gradient first, then $y - y_1 = m(x - x_1)$
parallel	same m
perpendicular	flip it and change the sign
do these three lie in a line?	gradient AB = gradient BC

Watch out. Almost none of this box is on your official formula sheet — only distance and midpoint are. The gradient formula and $m_1m_2 = -1$ are not.

Logs and exponentials

On the cheat sheet: **side 1 → LOGS AND EXPONENTIALS**

Find x (2 marks)

$5^x = 8$

1 The x is stuck up in the power. None of the usual moves reach it up there. This is exactly the problem logs were invented for.

```
you cannot divide it down
```

2 Take a log of both sides. Any base will do, as long as it is the same on both sides — and your calculator has lg, so use that.

$\lg(5^x) = \lg 8$

Cheat sheet: row **$5^x = 8$**.

3 Now the useful law: a power INSIDE a log comes out to the FRONT. That is the whole move, and it is why x is suddenly reachable.

$x \cdot \lg 5 = \lg 8$

That law IS on your official sheet, as $\log_s(m^n) = n \log_s m$.

4 It is an ordinary equation now. Divide both sides by lg 5.

$x = \lg 8 \div \lg 5$

5 Into the calculator, two decimal places.

$x = 1.29$

ANSWER $x = 1.29$

Evaluate (2 marks)

$\log_2 10$

1 Your calculator has a button for base 10 and one for base e. It has NO button for base 2, so you cannot type this in as it stands.

```
no base-2 button exists
```

2 **Change of base is not on your official sheet.** It is on the cheat sheet, in red, because you have to supply it yourself.

$\log_s x = \lg x \div \lg b$

Cheat sheet: row **$\log_2 10$, no button** — in red.

3 The number you want goes on top, the base goes underneath.

$\log_2 10 = \lg 10 \div \lg 2$

4 lg 10 is exactly 1, because ten to the power one is ten.

$= 1 \div 0.30103$

5 Divide.

$= 3.32$

ANSWER 3.32

A log is a power question asked backwards. Same three numbers, moved about.

$5^x = 8$ (power form) — same / fact — $x = \log_5 8$ (log form)

The base (the 5) stays the base. It never moves.

Your calculator only has TWO log buttons

`lg (base 10)` `ln (base e)` anything else, e.g. $\log_2$, has no button: change the base yourself.

$\log_s x = \lg x \div \lg b$ wanted number on top, base underneath

This one is NOT on your official formula sheet.

A log and a power are the same fact written two ways. The base never moves.

Spotting it in three seconds

If the question looks like	Your first move
the unknown is in the power	take lg of both sides, bring it down
log with an odd base	change of base: lg wanted ÷ lg base
two logs added	multiply the insides
two logs subtracted	divide the insides
ln appears	same rules, base e instead of 10

Watch out. The five log laws are on your official sheet. Change of base is NOT, and neither is the swap between $y = b^x$ and $x = \log_s y$.

Partial fractions, and the factor theorem

On the cheat sheet: **side 1 → PARTIAL FRACTIONS AND FACTOR THEOREM**

Express in partial fractions
4 marks

`(2x − 1) / ((2x − 3)(x − 1))`

1. Ignore the top completely for now. Look ONLY at the bottom, because its shape decides the shape of your answer.

   `two different brackets`

   Cheat sheet: row **two different brackets** in the shape table.

2. Two different brackets means one plain letter over each. Write this down before doing any arithmetic.

   `A/(2x−3) + B/(x−1)`

3. Multiply everything by the whole bottom. The brackets cancel one at a time and you are left with a plain equation.

   `2x − 1 = A(x−1) + B(2x−3)`

4. Now the trick that saves all the algebra: choose an x that KILLS one letter. Put x = 1, and the A bracket becomes zero.

   `2(1)−1 = B(2−3) → 1 = −B → B = −1`

5. Do it again with the x that kills B, which is x = 1.5.

   `2(1.5)−1 = A(0.5) → 2 = 0.5A → A = 4`

ANSWER **4/(2x − 3) − 1/(x − 1)**

Find a
2 marks

`(x − 3) is a factor of f(x) = x² + ax − 15`

1. A factor divides in exactly, leaving nothing behind. So if you feed in the x that makes the bracket zero, the whole thing must come to zero.

   `x − 3 = 0 when x = 3`

   Cheat sheet: row **is (x−k) a factor?**

2. **Watch the sign.** The bracket is (x − 3), so the number you use is PLUS 3. For (x + 6) you would use minus 6.

   `use x = 3`

3. Put 3 in everywhere, and set the whole lot equal to zero.

   `3² + 3a − 15 = 0`

4. Work out the numbers you can.

   `9 + 3a − 15 = 0 → 3a − 6 = 0`

5. Solve as normal.

   `a = 2`

ANSWER **a = 2**

Look at the BOTTOM only. Its shape decides the shape of your answer.

Bottom		Answer shape	
(x − 1)(x + 8) — two different brackets	→	A/(x−1) + B/(x+8)	one letter over each
(x + 4)² — the same bracket twice	→	A/(x+4) + B/(x+4)²	the bracket AND its square
x(x² + x + 7) — one has an x^2 that will not factorise	→	A/x + (Bx + C)/(x²+x+7)	that one gets Bx + C on top

Factor theorem in one line: to test (x − 2), put x = 2 in. Answer 0 means it IS a factor.
The sign flips: (x + 6) is tested with x = −6.

The bottom of the fraction tells you the answer shape. Pick the row that matches before you write anything.

Spotting it in three seconds

If the question looks like	Your first move
(x−1)(x+8)	A/(x−1) + B/(x+8)
(x+4)²	A/(x+4) + B/$(x+4)^2$
x(x²+x+7)	A/x + (Bx+C)/(x^2+x+7)
is (x−k) a factor?	put x = k in. Zero means yes

Watch out. The three shapes ARE on your official sheet. Which one to pick, and the kill-a-letter substitution, are not. Repeated brackets need BOTH powers.

Trigonometry: sides and angles

On the cheat sheet: **side 2 → TRIG — WHICH RATIO OR RULE?**

Find h
2 marks

`right-angled triangle, angle 40°, sloping side 8`

1. Label the three sides from the 40° corner first. Do this before anything else — most lost marks here are a mislabel.

   `8 is the SLOPING side = hypotenuse`

   Cheat sheet: **TRIG — WHICH RATIO OR RULE?**, the SOH CAH TOA strip.

2. h is the side straight across from the 40° corner, so h is the opposite.

   `h = opposite, 8 = hypotenuse`

3. Opposite and hypotenuse together is the S of SOH. So it is sine, not tan.

   `sin 40° = h ÷ 8`

   Cheat sheet: row **opposite and hypotenuse** → SOH.

4. Multiply both sides by 8 to get h on its own.

   `h = 8 × sin 40°`

5. Calculator in DEGREES, not radians. Check the little D on the screen.

   `h = 8 × 0.6428 = 5.14`

ANSWER `h = 5.14`

Find angle B
3 marks

`triangle ABC, C = 35°, AB = 40 cm, AC = 50 cm`

1. No right angle is mentioned, so SOH CAH TOA does not apply at all. It has to be one of the two rules.

   `no right angle → sine or cosine rule`

   Cheat sheet: row **no right angle** — both rules are in red.

2. Sides are named by the angle OPPOSITE them. AB is across from C, and AC is across from B. So you have two complete side-and-angle pairs.

   `AB = 40 pairs with C = 35°`

3. A complete pair plus one more side means the SINE rule. The cosine rule is for when you have no complete pair.

   `sin B / AC = sin C / AB`

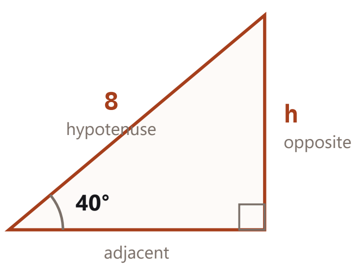
4. Put the numbers in and work out the right-hand side.

   `sin B = 50 × sin 35° ÷ 40 = 0.7167`

5. Now undo the sine with the inverse button.

   `B = sin⁻¹(0.7167) = 45.80°`

ANSWER `B = 45.80°`



The 8 lies along the SLOPE, so it is the hypotenuse.

Which of the three? Cover the one you want.

SOH	`sin = opp ÷ hyp`	you have the sloping side
CAH	`cos = adj ÷ hyp`	you have the sloping side
TOA	`tan = opp ÷ adj`	no sloping side involved

No right angle in the picture? Then it is a rule, not SOH CAH TOA.

$a/\sin A = b/\sin B$ a side and its OPPOSITE angle, paired

$a^2 = b^2 + c^2 - 2bc \cos A$ three sides, or two and the angle BETWEEN

Neither is on your official sheet. Both are on the cheat sheet.

Spotting it in three seconds

If the question looks like	Your first move
`right angle in the picture`	SOH CAH TOA
`a side and its opposite angle`	sine rule
`three sides, or two and the angle between`	cosine rule
`answer is an angle`	use the inverse button at the end

Watch out. Sine rule, cosine rule and SOH CAH TOA are all missing from your official sheet. They are on the cheat sheet in red. Check the calculator is in degree mode.

The 8 lies along the slope, so it is the hypotenuse. Using tan here gives 6.71, which is the usual wrong answer.

Trig equations, and reading a wave

On the cheat sheet: **side 2 → TRIG EQUATIONS — ALWAYS TWO ANSWERS, and WAVES**

Solve for θ

3 marks

$4 \sin \theta + 5 = 3$, for $0 \le \theta \le 360°$

1. Do not think about angles yet. Get sin θ on its own exactly as if it were an ordinary letter.

   $4 \sin \theta = 3 - 5 = -2$

2. Divide by 4.

   $\sin \theta = -0.5$

3. Press inverse sine. The calculator gives −30, which is outside the range asked for — that is normal, and it is not your answer yet.

   basic angle = 30°

   Cheat sheet: row **sin θ = negative**.

4. Sine is negative in the third and fourth quarters. Take the basic 30 and place it in each.

   $180 + 30 = 210$ and $360 - 30 = 330$

5. Both are inside 0 to 360, so both count. **Two marks, two answers.**

   $\theta = 210°, 330°$

ANSWER $\theta = 210°$ and $330°$

Find the equation

3 marks

sine graph, max 4, min −2, 2 cycles in 0–360, no shift

1. Four letters, and each is answered by one sentence of the question. Take them one at a time; do not try to see it all at once.

   $y = A \sin(B\theta + C) + D$

   Cheat sheet: row **A sin(Bθ+C)+D** — the four-letter table.

2. D is the middle line: average the top and the bottom.

   $D = (4 + (-2)) \div 2 = 1$

3. A is the height above that middle line: half the gap from top to bottom.

   $A = (4 - (-2)) \div 2 = 3$

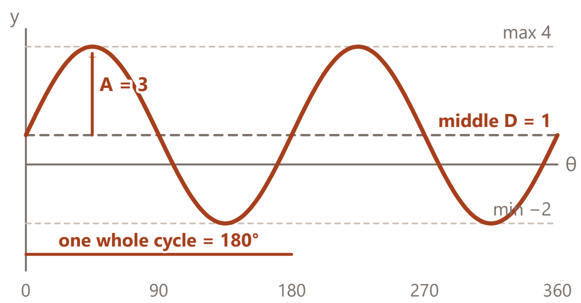
4. B is how many cycles fit into 360°. It says two.

   $B = 2$

5. "No shift" means C is nothing. Now put the four numbers back into the shape.

   $C = 0$

ANSWER $y = 3 \sin(2\theta) + 1$



$y = A \sin(B\theta + C) + D$

A	height	(max − min) ÷ 2
D	middle line	(max + min) ÷ 2
B	how many cycles	cycles in 360°
C	sideways shift	usually 0 — check the wording

sin θ = −0.5 gives TWO answers in 0–360
calculator says −30. Use 180+30 = 210 and 360−30 = 330.

Spotting it in three seconds

If the question looks like	Your first move
sin θ = positive	basic, and 180 − basic
sin θ = negative	180 + basic, and 360 − basic
cos θ = negative	180 − basic, and 180 + basic
tan θ = negative	180 − basic, and 360 − basic

Watch out. The identities are on your official sheet. Getting from one calculator answer to both real answers is not — that is the quadrant table on the cheat sheet.

The answer to the right-hand question, plotted. Middle line at 1, reaching 3 above and 3 below, twice over.

Vectors: size, direction and joining points

On the cheat sheet: **side 2 → VECTORS, and VECTOR DIRECTION — PLACE THE ANGLE**

Magnitude and direction (3 marks)

`the column vector [-7 ; -11]`

1. Two separate jobs in one question. Size first, direction second — never both at once.

   `size, then direction`

2. Size is Pythagoras. Square both, add, square root. The minus signs disappear when you square them, which is why size is never negative.

   $\sqrt{(-7)^2 + (-11)^2} = \sqrt{170} = 13.04$

   Cheat sheet: row **magnitude**.

3. For the direction, ignore the signs for a moment and get the basic angle from tan inverse.

   $\tan^{-1}(11 \div 7) = 57.53°$

4. **Now put the signs back.** Both parts are negative, so the arrow points down and to the left — the bottom-left quarter. The calculator will not do this for you.

   `bottom left → 180 + basic`

   Cheat sheet: the **quadrant** row, in red.

5. Add.

   $180 + 57.53 = 237.53°$

ANSWER 13.04 at 237.53°

Find vector AB (2 marks)

`A (2, 7, 2) and B (8, 4, 1)`

1. AB means the journey FROM A TO B. It is the finish take away the start.

   `AB = B − A`

   Cheat sheet: row **AB from two points**, in red.

2. **The letters read forwards but the sum works backwards.** AB is B minus A, not A minus B. Getting this the wrong way round flips every sign.

   `second letter first`

3. Take them one row at a time. Top row: 8 take away 2.

   $8 - 2 = 6$

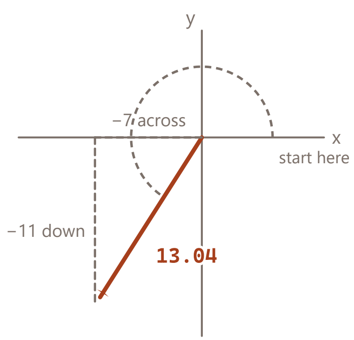
4. Middle row, then bottom row. Watch the second one: four take away seven is negative.

   $4 - 7 = -3 \quad 1 - 2 = -1$

5. Stack them back up as a column.

   `AB = [6 ; -3 ; -1]`

ANSWER AB = [6 ; -3 ; -1]



anticlockwise from the positive x axis → 237.53°

Two steps, always in this order

$\text{size} = \sqrt{7^2 + 11^2} = \sqrt{170} = 13.04$
the minus signs vanish, because they get squared

$\text{basic angle} = \tan^{-1}(11 \div 7) = 57.53°$
both negative → bottom left → 180 + 57.53
the calculator will NOT do this bit for you

Where the arrow points, and what to add

right and up	**the basic angle**
left and up	**180 − basic**
left and down	**180 + basic**
right and down	**360 − basic**

Spotting it in three seconds

If the question looks like	Your first move
magnitude, size, length, |a|	square, add, square root
direction or bearing	tan inverse, then fix the quarter
AB from points A and B	B minus A
resultant of two vectors	add them, then take the size

Watch out. Direction is measured anticlockwise from the positive x axis unless the question says otherwise. The calculator only ever gives you the basic angle.

Both parts negative puts the arrow in the bottom-left quarter, so the basic 57.53° becomes 237.53°.

Vectors: dot product and cross product

On the cheat sheet: **side 2 → VECTORS, dot and cross rows**

Angle between them

3 marks

`[12 ; 5] and [3 ; 7]`

1. The word ANGLE is the whole signal. Angle always means dot product, never cross.

```
angle → dot product
```

Cheat sheet: row **angle between two vectors**.

2. Dot product: multiply the tops together, multiply the bottoms together, then ADD. The answer is one plain number.

```
12×3 + 5×7 = 36 + 35 = 71
```

3. Now the size of each one separately. The first is a neat 13.

```
|a| = √169 = 13     |b| = √58 = 7.616
```

4. The formula divides the dot product by both sizes. **Forgetting to divide is the usual slip.**

```
cos θ = 71 ÷ (13 × 7.616) = 0.7169
```

5. Undo the cos with the inverse button.

```
θ = 44.18°
```

ANSWER `θ = 44.18°`

Find a × b

3 marks

`a = [1 ; 3 ; 0] and b = [0 ; 4 ; -1]`

1. Cross product is three dimensions only, and unlike the dot product the answer is another VECTOR, not a number.

```
answer will have three rows
```

2. For the TOP row, cover the top row of both and cross-multiply the four numbers left: 3×(−1) then take away 0×4.

```
(3)(−1) − (0)(4) = −3
```

Cheat sheet: row **cross product**, the cover-a-row method.

3. **The middle row is the odd one out.** Cover the middle row, then swap the order round: it is 0×0 minus 1×(−1), not the other way.

```
(0)(0) − (1)(−1) = 1
```

4. Bottom row, back to the normal order.

```
(1)(4) − (3)(0) = 4
```

5. Stack the three answers up.

```
a × b = [−3 ; 1 ; 4]
```

ANSWER `[−3 ; 1 ; 4]`

DOT a · b

multiply matching parts, then ADD them up

`[12; 5] · [3; 7] = 12×3 + 5×7 = 71`

The answer is ONE number. No direction.

cos θ = (a · b) ÷ (|a| × |b|)
use it whenever the question says ANGLE

`= 71 ÷ (13 × 7.616) = 0.7169`

θ = cos⁻¹ 0.7169 = 44.18°

Forgetting to divide by the two sizes is the usual slip.

CROSS a × b

three dimensions only. The answer is a VECTOR.

`a = [1; 3; 0]   b = [0; 4; -1]`

i	cover row 1	3×(−1) − 0×4	→ **−3**
j	cover row 2	0×0 − 1×(−1)	→ **1**
k	cover row 3	1×4 − 3×0	→ **4**

the middle one (j) is the other way round

Spotting it in three seconds

If the question looks like	Your first move
`angle between`	dot product, then divide by both sizes
`are they perpendicular?`	dot product comes to zero
`perpendicular VECTOR wanted`	cross product
`answer should be one number`	dot. A column? cross

Watch out. Both formulas are on your official sheet. Knowing which one the question is asking for is not — and neither is the sign flip on the middle row.

Dot gives a number and answers ANGLE questions. Cross gives a vector, and its middle row runs backwards.

Matrices

On the cheat sheet: **side 2 → MATRICES**

Solve using the inverse

4 marks

`[3  7 ; 5  9] [x ; y] = [16 ; −24]`

1. This is already in the standard shape: a matrix times the unknowns equals the answers. Name the three parts and it stops looking strange.

   `M U = V, so U = M⁻¹ V`

   Cheat sheet: row **solve with a matrix**, in red.

2. The determinant comes first, because if it were zero there would be no inverse and no answer. Top-left times bottom-right, take away the other diagonal.

   `det = 3×9 − 7×5 = 27 − 35 = −8`

3. To invert a 2×2: swap the two on the main diagonal, flip the signs of the other two, then divide the lot by the determinant.

   `M⁻¹ = (1/−8) [9  −7 ; −5  3]`

4. **Order matters here.** The inverse goes on the LEFT of the answers column. $M^{-1}V$ and VM^{-1} are not the same thing.

   `[x ; y] = (1/−8) [9  −7 ; −5  3] [16 ; −24]`

5. Row times column: 9×16 + (−7)×(−24) = 312, then divide by −8. Same again for the second row.

   `x = 312 ÷ −8 = −39    y = −152 ÷ −8 = 19`

ANSWER `x = −39, y = 19`

State the order of the product

2 marks

`[1 ; 2 ; −1]  ×  [0  0  3  −2]`

1. Write down the size of each one first, rows before columns, always in that order.

   `first is 3 × 1, second is 1 × 4`

   Cheat sheet: row **order of a product**.

2. Write the two sizes side by side and look at the two numbers that end up touching in the middle.

   `(3 × 1)(1 × 4)`

3. The two INSIDE numbers must be equal or it cannot be done at all. Here they are both 1, so it is legal.

   `1 = 1 ✓`

4. The answer takes the two OUTSIDE numbers.

   `outside: 3 and 4`

5. So the product is a 3-row, 4-column matrix.

   `3 × 4`

ANSWER `3 × 4`

Before multiplying: check the two INSIDE numbers match.

`2 × 3` `3 × 4` → **`2 × 4`**

the OUTSIDE two

these two must be equal — if not, it cannot be done

Solving with a matrix: M U = V, so U = M^{-1} V

$$\underbrace{\begin{bmatrix}3 & 7\\5 & 9\end{bmatrix}}_{M} \times \underbrace{\begin{bmatrix}x\\y\end{bmatrix}}_{U} = \underbrace{\begin{bmatrix}16\\-24\end{bmatrix}}_{V}$$

`det = 3×9 − 7×5 = −8`

swap the corners, flip the signs of the other two, then divide everything by the determinant.

det = 0 → no inverse → cannot be solved this way

Inside numbers must match, outside numbers give the answer. One glance tells you whether a multiplication is even allowed.

Spotting it in three seconds

If the question looks like	Your first move
`state the order`	rows × columns. Inside must match
`solve for x and y`	U = M^{-1} V
`find the inverse`	swap, flip, divide by det
`det = 0`	no inverse. Say so and stop

Watch out. Determinant and inverse formulas are on your official sheet. That the ORDER of multiplication matters, and that det = 0 means no solution, are not.

Differentiation: the rules

On the cheat sheet: **side 2 → DIFFERENTIATE — READ THE FUNCTION**

Find dy/dx 3 marks

$y = 4x^5 + 7x + 6/x$

1. Three terms, so three separate little jobs. Do them one at a time and write each answer down before starting the next.

   `deal with each term alone`

2. First term: the power comes down to the front and multiplies, then the power itself drops by one.

   $4x^5 \rightarrow 5\times4x^4 = 20x^4$

   Cheat sheet: row **x^n** in the derivative table.

3. Second term: $7x$ is really $7x^1$, and the 1 comes down leaving x^0, which is 1. So a plain x always differentiates to its number.

   $7x \rightarrow 7$

4. **Rewrite the third one before touching it.** An x on the bottom is a negative power. This is the step people skip.

   $6/x = 6x^{-1}$

   Cheat sheet: the **rewrite first** note, in red.

5. Now the same rule: bring down the −1, and the power goes to −2. Minus one times six is minus six.

   $-6x^{-2} = -6/x^2$

ANSWER $dy/dx = 20x^4 + 7 - 6/x^2$

Find dy/dx 3 marks

$y = \ln(x^4 + 4x)$

1. This is a log with a whole expression inside it, not just x. There is one rule for exactly this and it is worth knowing by heart.

   $d/dx \ln f(x) = f'(x) / f(x)$

   Cheat sheet: row **ln f(x)**, in red.

2. Name the inside so you stop having to look at the whole thing.

   the inside is $x^4 + 4x$

3. Differentiate ONLY the inside, on its own, ignoring the ln entirely.

   $f'(x) = 4x^3 + 4$

4. Now assemble: what you just worked out goes on top, and the untouched inside goes underneath.

   $(4x^3 + 4) / (x^4 + 4x)$

5. You can leave it exactly like that. Tidying is optional and risks a slip.

   `done`

ANSWER $dy/dx = (4x^3 + 4) / (x^4 + 4x)$

The power rule, in pictures

$x^n \xrightarrow{\text{differentiate}} n\,x^{n-1}$

$6/x$ is $6x^{-1}$ — rewrite it BEFORE you start

the power comes DOWN to the front, then goes down by one

The four that are not powers

$\sin(ax)$	$a\cos(ax)$	sin becomes cos, and the a drops out front
$\cos(ax)$	$-a\sin(ax)$	cos becomes MINUS sin
e^{ax}	$a\,e^{ax}$	it barely changes, which is why e is special
$\ln f(x)$	$f'(x) \div f(x)$	differentiate the inside, put it over the inside

Chain rule: the a in front comes from the inside. Nothing else does.

The power rule, and the four that do not follow it. Anything on the bottom of a fraction gets rewritten first.

Spotting it in three seconds

If the question looks like	Your first move
`x to a power`	power to the front, then take one off it
`something / xⁿ`	rewrite as a negative power FIRST
`e to something`	copy it, times the derivative of the power
`ln(something)`	inside differentiated, over the inside
`two brackets multiplied`	expand them first, then power rule

Watch out. The standard derivatives are on your official sheet. The rewrite step, and the chain rule for what is inside a bracket, are not.

Turning points, and word problems

On the cheat sheet: **side 1 → TURNING POINTS, AND WORD PROBLEMS**

Turning point and its nature
4 marks

$y = 5 + 6x - 3x^2$

1. A turning point is where the curve stops climbing and starts falling, so for one instant it is flat. Flat means the gradient is zero.

   flat → $dy/dx = 0$

   Cheat sheet: row **turning point** — three steps, in order.

2. Differentiate. The 5 on its own disappears, because a constant has no slope.

   $dy/dx = 6 - 6x$

3. Set that to zero and solve. This gives you the x, and only the x.

   $6 - 6x = 0 \to x = 1$

4. **Put x back into the ORIGINAL equation**, never into the differentiated one, to get the y that goes with it.

   $y = 5 + 6(1) - 3(1) = 8$

5. Nature: differentiate a second time. Negative means the curve is a frown, so the point is the highest.

   $d^2y/dx^2 = -6$, negative → MAXIMUM

ANSWER (1, 8), a maximum

Time to maximum height
3 marks

$h(t) = -5t^2 + (v \sin\theta)t$, $v = 100$ m/s, $\theta = 30°$

1. It looks frightening because of the letters. Work out the number bit first and most of the fright goes.

   $100 \times \sin 30° = 100 \times 0.5 = 50$

2. Rewrite it with that number in place. It is now an ordinary quadratic.

   $h = -5t^2 + 50t$

3. "Maximum height" is a turning point, so the same three steps apply. Differentiate with respect to t.

   $dh/dt = -10t + 50$

   Cheat sheet: the same **turning point** row.

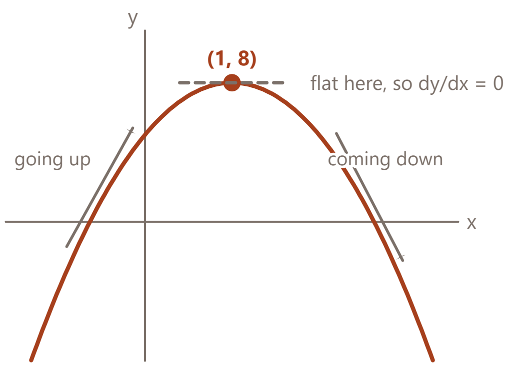
4. At the very top it is momentarily neither rising nor falling, so set that to zero.

   $-10t + 50 = 0 \to t = 5$

5. Read the question again. It asked for the TIME, so 5 seconds is the answer — do not go on and work out the height unless asked.

   $t = 5$ s

ANSWER t = 5 seconds



The curve from the left-hand question. Flat at the top, climbing before it, falling after — so that point is a maximum.

Three steps, every single time

1. differentiate — $dy/dx = 6 - 6x$
2. set it to zero, solve — $6 - 6x = 0 \to x = 1$
3. put x back in the ORIGINAL — $y = 5 + 6 - 3 = 8$

Then say which kind it is

d^2y/dx^2 negative	d^2y/dx^2 positive
MAXIMUM ∩	MINIMUM ∪

Here it is −6, which is negative, so (1, 8) is a maximum.

Spotting it in three seconds

If the question looks like	Your first move
turning point, stationary point	dy/dx = 0
maximum or minimum value	same, then second derivative
rate of change at t = 3	differentiate, then put 3 in
largest area / volume	differentiate, set to zero
when does it stop rising?	the derivative is zero

Watch out. Nothing in this box is on your official sheet. Second derivative negative means maximum — that is backwards from what most people guess.

Integration: the rules

On the cheat sheet: **side 2 → INTEGRATE — READ THE INTEGRAND**

Find the integral (3 marks)

$\int 7/(3 - 4x)\,dx$

1 A number over a bracket with x in it. This is the one case where the power rule does NOT work, and the answer is a log instead.

$\int a/(bx + d)\,dx = (a/b) \ln|bx + d| + c$

Cheat sheet: row **a/(bx+d)**, in red.

2 Name the parts so the formula has somewhere to put them. The b is the number in front of x, and here it is negative.

$a = 7 \quad b = -4 \quad d = 3$

3 The number out front is a divided by b.

$7 \div -4 = -1.75$

4 The bracket goes inside a log, with the straight lines around it.

$-1.75 \ln|3 - 4x|$

5 **Add the + c.** There are no limits on that integral sign, so the constant is worth a mark and it is the easiest mark on the paper.

$-1.75 \ln|3 - 4x| + c$

ANSWER $-1.75 \ln|3 - 4x| + c$

Find y (3 marks)

$dy/dx = 4x^3 - 6x^2 + x$, and $y = 3$ when $x = 0$

1 You are given the gradient and asked for the curve, which is the backwards direction. Backwards means integrate.

`integrate each term`

2 Integrating is the opposite of differentiating: the power goes UP by one, then you divide by that new power.

$4x^3 \to 4x^4/4 = x^4$

Cheat sheet: row $\mathbf{x^n}$ in the integral table.

3 Same for the other two. The lone x becomes x^2 divided by 2.

$-6x^2 \to -2x^3 \qquad x \to x^2/2$

4 So far, plus the constant you do not know yet.

$y = x^4 - 2x^3 + x^2/2 + c$

5 Now use the fact you were given. Put x = 0 and y = 3 in: everything with an x vanishes, and c is left standing on its own.

$3 = 0 - 0 + 0 + c \to c = 3$

ANSWER $y = x^4 - 2x^3 + x^2/2 + 3$

---

differentiate: power down to the front

$x^4 \rightleftarrows 4x^3$

integrate: power UP by one, then divide by the new power

They undo each other. If you are unsure, differentiate your answer back.

The three that do not follow the power rule

$\int a/(bx + d)\,dx$	$(a/b) \ln|bx + d| + c$	divide by the number in front of x
$\int \sin(ax)\,dx$	$-(1/a)\cos(ax) + c$	a minus appears from nowhere
$\int e^{ax}\,dx$	$(1/a)\,e^{ax} + c$	divide by the a

No limits on the integral sign? Then + c is a mark. It is the easiest mark on the paper and the most often dropped.

Integration is differentiation run backwards. If you are ever unsure, differentiate your answer and see if you get back.

Spotting it in three seconds

If the question looks like	Your first move
`x to a power`	add one to the power, divide by the new one
`number / (bracket with x)`	log. Divide by the number in front of x
`sin(ax)`	minus cos, divided by a
`e to something`	copy it, divide by the number in front of x
`given a point on the curve`	integrate, then put the point in to find c

Watch out. The standard integrals are on your official sheet. The a/(bx+d) log case is not, and neither is the reminder that no limits means you owe a + c.

Definite integrals, and area

On the cheat sheet: **side 2 → INTEGRATE — READ THE INTEGRAND, area rows**

Area of the shaded region

4 marks

$y = 5x^3 - 2x + 8$, between $x = 1$ and $x = 3$

1. Area under a curve between two x values is a definite integral. Write it out with the limits on the sign before doing anything.

   $\int$ from 1 to 3 of $(5x^3 - 2x + 8)\,dx$

   Cheat sheet: row **area under a curve**.

2. Integrate exactly as normal, term by term. **No + c this time** — with limits it would cancel itself out anyway.

   $[\ 5x^4/4 - x^2 + 8x\ ]$

3. Put the TOP number in first. Every x becomes a 3.

   $5(81)/4 - 9 + 24 = 101.25 - 9 + 24 = 116.25$

4. Now the bottom number, into the same expression. Every x becomes a 1.

   $5/4 - 1 + 8 = 8.25$

5. Top answer take away bottom answer, in that order. Swapping them gives the right number with the wrong sign.

   $116.25 - 8.25 = 108$

ANSWER **area = 108**

Evaluate

3 marks

$\int$ from 1 to 3 of $6e^{2x}\,dx$

1. An e with a number in front of the x. Integrating divides by that number, and here it is 2.

   $\int e^{ax}\,dx = e^{ax}/a$

   Cheat sheet: row **e^{ax}** in the integral table.

2. The 6 out front just comes along for the ride. Six divided by two is three.

   $[\ 6e^{2x}/2\ ] = [\ 3e^{2x}\ ]$

3. Top limit first: x = 3 makes the power 6.

   $3e^6 = 3 \times 403.43 = 1210.29$

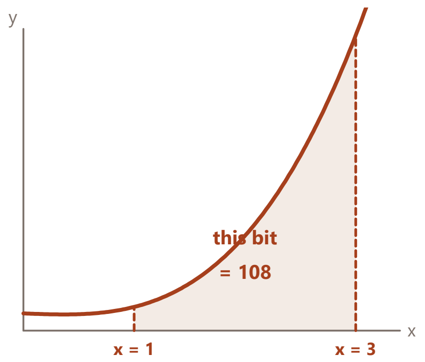
4. Then the bottom: x = 1 makes the power 2.

   $3e^2 = 3 \times 7.389 = 22.17$

5. Subtract, top minus bottom, and round to two decimal places as the paper asks.

   $1210.29 - 22.17 = 1188.12$

ANSWER **1188.12**



The shaded strip is what the definite integral measures. The limits are the two vertical edges.

Definite integral: the same working, then two sums

1. integrate as normal, and drop the + c
2. square brackets, limits on the right
3. put the TOP number in
4. put the BOTTOM number in
5. top answer − bottom answer

$[\ 5x^4/4 - x^2 + 8x\ ]$ from 1 to 3
$= 116.25 - 8.25 = 108$

If the curve dips BELOW the axis
split it there, and make the negative bit positive.

Spotting it in three seconds

If the question looks like	Your first move
numbers on the integral sign	definite. No + c
area bounded by, shaded region	definite integral between those x values
the curve dips below the axis	split it there, make the negative bit positive
area between two curves	top curve minus bottom curve, then integrate

Watch out. Splitting at the axis is not on your official sheet, and it is the one that quietly costs marks — the negative part cancels the positive part if you do not.

First order differential equations

On the cheat sheet: **side 1 → FIRST ORDER DIFFERENTIAL EQUATIONS — MATCHING ONLY**

Set up the equation
3 marks

`coffee at 90°C in a 20°C room, cooling rate constant k = 0.01`

1. Translate the sentence one phrase at a time. "The rate at which the temperature changes" is dT/dt. That is all that phrase ever means.

   `rate of change → dT/dt`

   Cheat sheet: row **cooling towards a room**, in red.

2. "Proportional to the difference between its temperature and the room" gives you the bracket. Difference means subtract.

   $(T - 20)$

3. **The coffee is cooling, so it must be a minus.** Without the minus sign the equation says the coffee heats up forever.

   $dT/dt = -k(T - 20)$

4. Put the given k in.

   $dT/dt = -0.01(T - 20)$

5. The solution has the shape: room temperature plus a shrinking bit. At t = 0 the coffee is 90, so the shrinking bit starts at 70.

   $T = 20 + 70e^{-0.01t}$

ANSWER $dT/dt = -0.01(T - 20), \quad T = 20 + 70e^{-0.01t}$

Time to reach room temperature
2 marks

`from the graph, estimate when the coffee reaches 20°C`

1. Try it algebraically first and watch what happens. Set T to 20 and see where it leads.

   $20 = 20 + 70e^{-0.01t}$

2. Take the 20 off both sides. You are left with something that can never be true, because e to any power is never zero.

   $0 = 70e^{-0.01t}$ — impossible

3. **So it never actually gets there.** The curve gets closer and closer to 20 forever without touching. That is what that flat dashed line in the picture means.

   `it approaches 20, never reaches it`

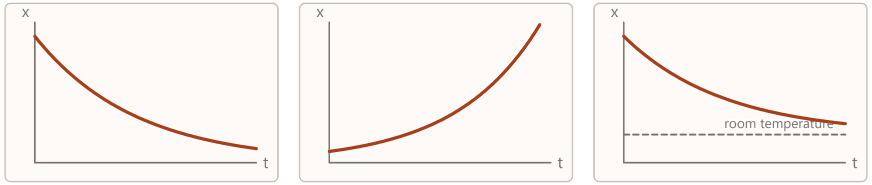
4. Which is why the question says ESTIMATE FROM THE GRAPH. Put a big number in and see how close it is by then.

   $T(500) = 20 + 70e^{-5} = 20.47°$

5. By 500 minutes it is within half a degree, which on a graph is indistinguishable. Say so in words — that is the answer they want.

   `about 450 to 500 minutes`

ANSWER **roughly 450–500 min; strictly it never reaches 20°**



decay
$dx/dt = -kx$
$x = Ae^{-kt}$
falls, flattening towards zero

growth
$dx/dt = kx$
$x = Ae^{kt}$
rises, getting steeper

cooling
$dx/dt = -k(x - c)$
$x = c + Ae^{-kt}$
falls towards the room, never reaching it

The minus sign in front of k is the whole answer to the multiple choice.
Going down needs a minus. Cooling TOWARDS something needs the (x − c) bracket as well.

The three shapes the multiple-choice questions use. Match the picture to the minus sign and the bracket.

Spotting it in three seconds

If the question looks like	Your first move
`decays, halves, dies away`	$dx/dt = -kx \rightarrow x = Ae^{-kt}$
`grows, doubles, multiplies`	$dx/dt = kx \rightarrow x = Ae^{kt}$
`cools or warms TOWARDS something`	$dx/dt = -k(x - c)$
`levels off at a limit`	the answer has a + c in front of the e

Watch out. Nothing here is on your official sheet. If the quantity falls, k needs a minus in front of it — that single sign decides most of the multiple choice.